Containing Epidemics on the Global Air-Transport Network: Budget, Centrality, and the Epidemic Threshold

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TABLE I
OpenFlights network summary

Measure	Value
Airports (full graph)	3,214
Routes (full graph)	18,858
Largest-component airports	3,188
Largest-component routes	18,833
Average degree (largest component)	11.81
Connected components	7

Abstract—We study how limited airport closures affect epidemic spreading on the global air-transport network. On a cleaned OpenFlights graph of 3,188 airports, we compare random closure against degree-, closeness-, betweenness-, and eigenvector-targeted strategies under equal budgets of 1–20%, simulated with an Euler-integrated SIR metapopulation model. At a 20% budget, targeted strategies cut the final outbreak to 1.8–10.1%, versus 61.6% for random closure, but fragment the network: the strategy that best suppresses the epidemic is not the one that best preserves connectivity.

I. Motivation / Problem Statement

Air travel connects distant populations and can accelerate geographic epidemic spread. Since closing all airports is impractical, this work asks which airports should be closed under a limited budget. [1], [2]

II. Methodology

A. Network construction

We use the OpenFlights airport and route files downloaded on 27 October 2026 [3]. Airports are nodes and routes are undirected edges. Invalid numeric identifiers, self-loops, duplicate routes, and isolated airports are removed. The cleaned graph has 3,214 airports and 18,858 connections in seven components; its largest component has 3,188 airports and 18,833 connections. Table I summarizes the graph used for the epidemic experiments.

B. Centrality and interventions

We calculate degree, closeness, betweenness, and eigenvector centrality using NetworkX, NumPy, pandas, and Matplotlib [4], [5], [6], [7]. Degree counts direct connections; closeness measures average distance to all other airports; betweenness counts shortest paths passing through an airport; eigenvector rewards connections to other important airports. For a budget b, the top $\lceil b|V| \rceil$ airports are removed for a centrality strategy. Random closure samples the same number of airports using a fixed seed. Table II lists the top airports identified by each centrality measure.

TABLE II
Top airports by centrality in the largest component

Measure	Top three airports (OpenFlights ID)
Degree	Schiphol (580), Frankfurt (340), Charles de Gaulle (1382)
Closeness	Frankfurt (340), Charles de Gaulle (1382), Heathrow (507)
Betweenness	Charles de Gaulle (1382), LAX (3484), Anchorage (3774)
Eigenvector	Schiphol (580), Frankfurt (340), Charles de Gaulle (1382)

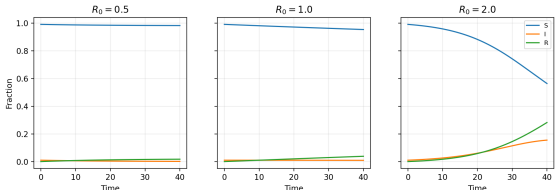


Fig. 1. Mean-field SIR behavior below, at, and above the epidemic threshold.

C. SIR threshold

The normalized mean-field model is

$$\dot{S} = -\beta SI, \quad \dot{I} = \beta SI - \gamma I, \quad \dot{R} = \gamma I. \tag{1}$$

The threshold is $R_0 = \beta S_0/\gamma$; with $\beta = 0.30$ and $\gamma = 0.10$, $R_0 = 3 > 1$, so an outbreak can grow. The disease-free equilibrium is stable below $R_0 = 1$ and loses stability at the transcritical threshold (Fig. 1).

D. Network metapopulation model

Each airport has susceptible, infected, and recovered populations. Local transmission is $\beta S_i I_i/N_i$, recovery is γI_i, and an undirected graph-diffusion term couples neighbouring airports. For each compartment $X_i \in \{S_i, I_i, R_i\}$, mobility adds $\mu\left(\sum_{j\in N(i)} X_j - k_i X_i\right)$, where k_i is airport i's degree. Euler integration is used with $\beta = 0.30$, $\gamma = 0.10$, mobility rate 0.02, normalized population one per airport, $dt = 0.1$, and duration 80. The initial infected

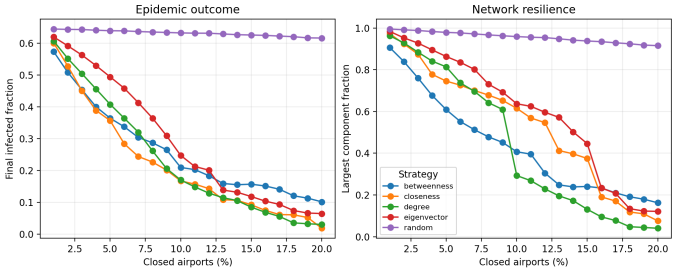


Fig. 2. Complete 1–20% closure-budget results on the largest OpenFlights component.

TABLE III
Final outbreak and connectivity at a 20% budget

Strategy	Final outbreak	Peak infected	LCC fraction
Random	61.63%	25.45%	91.55%
Degree	3.00%	0.80%	4.04%
Closeness	1.80%	0.59%	7.57%
Betweenness	10.11%	3.22%	16.24%
Eigenvector	6.43%	2.22%	12.12%

fraction is 1% at one fixed high-degree seed airport. The seed airport is protected from closure so that every strategy is evaluated from the same initial condition. For immediate closure, selected airports and their incident routes are removed before simulation. For delayed closure, incident mobility edges are removed and local infection at closed airports is set to zero at the intervention time; their existing populations remain and can recover.

E. Interventions and measurements

We compare random, degree, closeness, betweenness, and eigenvector closures for budgets from 1% to 20%. Outcomes include final outbreak size, peak infection, time to peak, and largest connected component. Final outbreak size is the cumulative recovered fraction at the end of the simulation. We define containment as final outbreak size at most 5%. The complete sweep evaluates every integer budget from 1% to 20%; random closure uses five independent trials.

III. Results and Key Findings

Figure 2 shows the complete generated real-network comparison. At a 20% budget, closeness has the smallest final outbreak size (1.80%), followed by degree (3.00%), eigenvector (6.43%), betweenness (10.11%), and random closure (61.63%). Degree reaches the 5% containment criterion first, at 18

The compact synthetic comparison was used as a short-horizon implementation check across ER, BA, and WS networks at a 10% closure budget. Because these validation runs ended at time 20 while infection was still active, they are not used for the main containment conclusion. The degree-preserving null comparison retained almost the same random-closure connectivity as the real network (96.5% versus 96.6

Delayed intervention was evaluated with a fixed 10% betweenness closure budget. With immediate, 20-unit, and 40-unit intervention, final outbreak sizes were 18.83%, 54.15%, and 57.02%, respectively, and peak infected fractions were 6.30%, 23.21%, and 24.44%. Thus, intervening later substantially increases outbreak size in this model.

IV. Discussion / Analysis

The real network is heterogeneous: a small set of airports has a large role in connecting otherwise weakly connected regions. Degree and closeness performed best for reducing the cumulative outbreak under the tested parameters, whereas betweenness caused strong fragmentation but did not achieve the 5% containment criterion by 20%. Anchorage ranks third on betweenness because many intercontinental shortest paths pass through high-latitude connections, yet it is not the most valuable target for suppressing transmission in this model: structural importance for connectivity and epidemic suppression need not be identical. The appropriate strategy depends on whether the objective prioritizes epidemic reduction, connectivity preservation, or a weighted combination of both.

The model has several limitations. OpenFlights route records do not provide passenger populations or time-varying traffic, so normalized airport populations and unweighted routes are used. The SIR parameters are scenario parameters rather than estimates for a particular disease. Finally, the intervention comparison measures structural removal and does not model rerouting or behavioral adaptation.

V. Conclusion

This project provides a reproducible framework[1] for comparing airport closure strategies on a real flight network. The mean-field model validates the epidemic threshold, while the metapopulation model connects local SIR dynamics to global network structure. Degree reached the 5% containment criterion first, at 18% closure, while closeness produced the smallest final outbreak at 20% (1.80%); both achieved this by strongly fragmenting the network, so effective control must be balanced against connectivity loss.

References

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